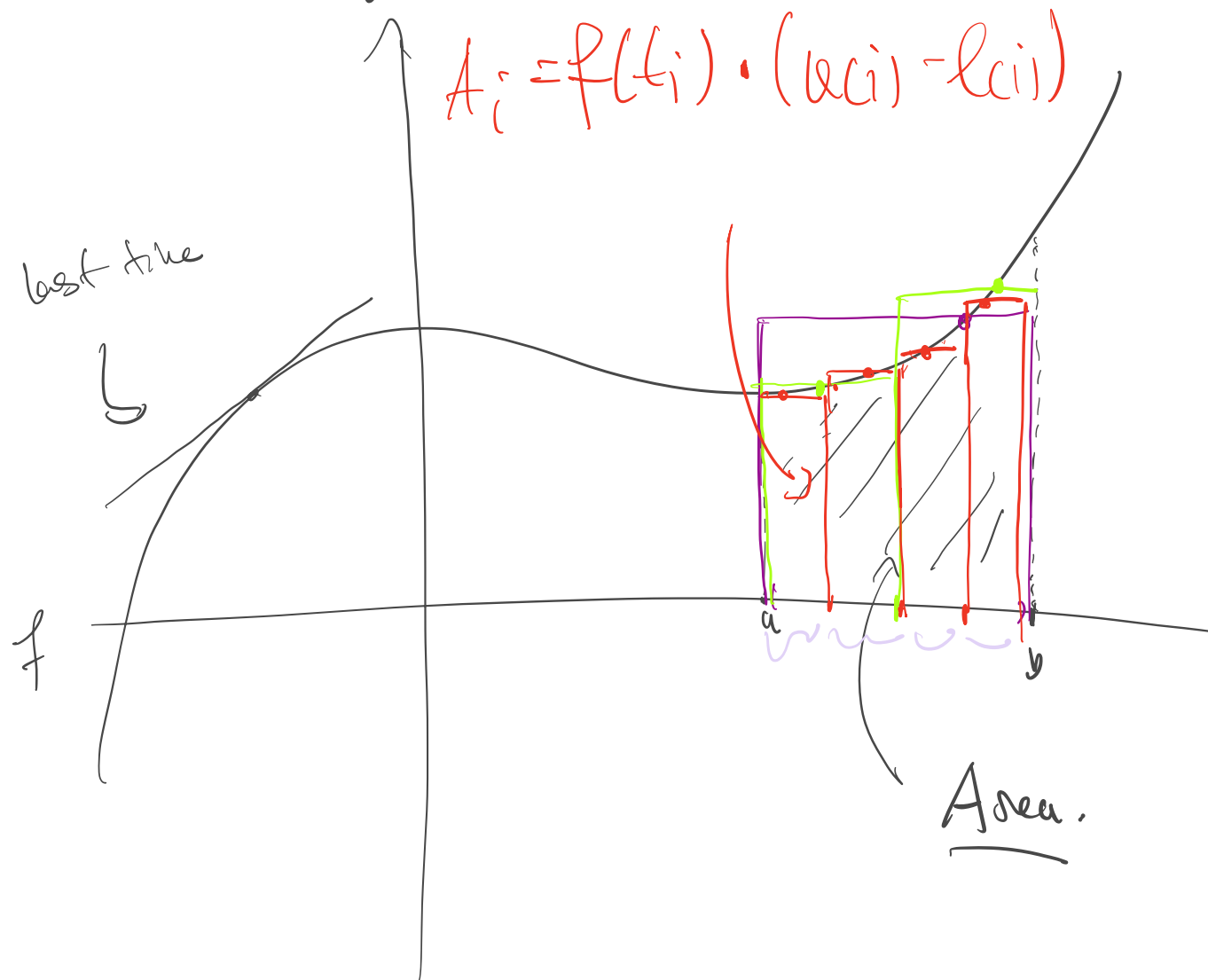


Lecture 9: Integration



Idea. To compute the area under the graph using slices that get thinner & thinner

Defn A partition of $[a, b]$ is the following:

- lower bounds $l : \text{Fin } n \rightarrow \mathbb{R}$
- upper bounds $u : \text{Fin } n \rightarrow \mathbb{R}$

such that

$$\bullet \quad l(0) = a ; \quad u(n-1) = b.$$

$$\bullet \quad l(i) < u(i) \quad \forall i \in \text{Fin } u.$$

$$\bullet \quad u(i) = l(i+1) \quad (i \neq 0, \neq n-1).$$

A tagged partition is a partition

with a seq of points

$$t: \text{Fin } u \rightarrow \mathbb{R}$$

$$\text{s.t.} \quad t(i) \in [l(i), u(i)]$$

Defn. Let $f: [a, b] \rightarrow \mathbb{R}$ and

(l, u, t) be a tagged partition

of $[a, b]$ then the Riemann sum

of f w.r.t. (l, u, t) is

$$RS(f, (l, u, t))$$

$$= \sum_{i=0}^{u-1} f(t(i)) \cdot (u(i) - l(i)),$$

Idea. The area under the graph of f over $[a, b]$ (the integral of f over $[a, b]$) should be the limit of these sums as we approximate it

with finer & finer
partitions!

Defn. We say a partition is subordinate
to $\delta > 0$ if $[u(i), l(i)] \subset (t(i) - \delta, t(i) + \delta)$
 $\forall i \in F_n$.

Defn. We say a function $f: [a, b] \rightarrow \mathbb{R}$
is integrable if $\exists A \in \mathbb{R}$ st.

$\forall \varepsilon > 0 \exists \delta > 0$ st.

$\forall$ tagged partitions subordinate to δ (l, u, t)

$$| A - R(f, (l, u, t)) | < \varepsilon.$$

Notation: If f is integrable we write
 $\int_a^b f(x) dx$ for this value A .

Problems 1) Such partitions actually exist!
2) Is this number A unique?

A: Yes!

We know that limits are unique.

If we can find a seq number
 x_n converging to every A that
satisfies the integrability cond
then we know A is unique!

For $n > 0$. Pick a partition

that satisfies

(ℓ, a, t)

$$|A - \text{RIS}(\ell, a, t)| < \frac{1}{n}.$$

x_n

does not depend on A .

and $A = \lim_{n \rightarrow \infty} x_n$.

How to compute the Riemann
integral?

Theorem, let $f: \mathbb{R} \rightarrow \mathbb{R}$ be diff.

and f' be integrable on $[a, b]$.

then $\int_a^b f'(x) dx = f(b) - f(a)$.

u

Rem. • The assumption that f' is
integrable is ~~messy~~!

- Continuous are always integrable!
-
-

How is it done in mathlib?

1) What kind of functions can you integrate?

$$f: \mathbb{R}^n \longrightarrow \cancel{\mathbb{R}} \quad \mathbb{E}$$

$\uparrow$
R-V.S.

Intervalls $\longrightarrow$ Boxes (products of intervals)

length $\longrightarrow$ volume

Riemann Sum becomes

$$\sum \underbrace{f(t_i)}_{\mathbb{E}} \cdot \underbrace{(\text{vol}(\text{Box}))}_{\mathbb{R}}$$

Next generalisation:

volume of a box:

$\text{vol}(\text{Box}) \in \mathbb{R} \xrightarrow{F}$ volume function

which is additive w.r.t.

unions of boxes,

Replacing the real volume
with more general volume

means. $\text{vol} : \text{Boxes} \times E \rightarrow F$
 $\uparrow \quad \uparrow$
add linear.

Proof of the fundamental theorem

theorem :

Mean value theorem :

on an interval $[a, b]$, f : diff.

$$f(b) - f(a) = f'(x_0)(b-a) \quad \text{for some } x_0 \in (a, b).$$

Observation: r.h.s. ✓

$$RS(f', (a, b), x_0)$$

We can generalise this observation:

Given any partition (l, u) ^(n terms)
of $[a, b]$ we can find
 $t(i) \in (l(i), u(i))$

sf. $f(u_i) - f(l_i) =$

$$f'(t(i)) (u(i) - l(i))$$

these are the
summed in RS,

Thus

$$RS(f'(l, u, t))$$

u-1

$$= \sum_{i=0}^n f'(t_i) (u_i - l_i)$$

$$= \sum_{i=0}^{n-1} f(u_i) - f(l_i)$$

telescope!

$$= f(b) - f(a)$$

To prove theorem we need
to do the following:

let $\epsilon > 0$

Then since f' is int. func

exists a $\delta > 0$ st.

$\forall$ tagged part. subord. to $\delta(t, \eta, t)$

$$\left| \int_a^b f' dx - R S(f', (t, \eta, t)) \right| < \epsilon$$

$$\left[\text{idea, } \underbrace{\hspace{10em}}_{= f(b) - f(a)} \right]$$

All we need is to find a
partition of $[a, b]$ subordinate
to δ .

There exists N st. $\frac{b-a}{N} < \delta$.

then define

$$l: \text{Fin } \mathbb{N} \rightarrow [a, b]$$

$$i \mapsto a + \frac{i(b-a)}{N}$$

$$u: \text{Fin } \mathbb{N} \rightarrow [a, b]$$

$$i \mapsto a + \frac{(i+1)(b-a)}{N}$$

Thus define a partition $\mathcal{Q}$

& tagging it is sub- to $\mathcal{S}$!

$$\Rightarrow \left| \int_a^b f' dx - (f(b) - f(a)) \right| < \varepsilon,$$

□

Difficulties for the formalization:

1) Defining the partition!

2) Telescoping

because in Lean

Hilbert Sum: is a sum

over an finite type

→ not ordered

How to approach them?

Answer: Use what's
already there!

[1) This kind of partition
for $[0, 1]$.

[2) A trick, similar
result in the
line

Library

"telescoping for Box Add
function".

How can we use

1) to construct

general partitions?

A: We would want a
result about how
partitions behave under
affine transformations.

$$\left(\begin{array}{l} \varphi: \mathbb{R} \rightarrow \mathbb{R} \\ x \mapsto ax + b \end{array} \right)$$

Since affine transformations
are invertible ($a \neq 0$),

such a φ takes

a partition of

$[a, 1]$ to a partition
of $[b, a+b]$. (if $a > 0$)

If a partition of $(0, 1)$

is subordinate to δ

then φ (partition) is

subordinate $\delta \circ \alpha$.